

Kernel Architecture Guide

not really an architecture, more so an explanation of how the kernels are meant to be modularized

`cuda_utils.cuh`: utility file for kernels, contains macros for error checking and testing

`*.cuh`: create one for each kernel, just like cpp files are laid out (ie cuda header file)

`*cu`: actual kernel, keep clean and concise, no testing

`test_*.cu`: one for each kernel, test kernel outputs against their cpu counterparts. basically the "main" block that is usually in kernel files

the kernel contract is basically:

input: flattened [B, N, D]

output: flattened [B, N, D, F, 2]

for each coordinate scalar x:

$\sin(\pi * 2^f * x)$

$\cos(\pi * 2^f * x)$